Three layers. One accountable owner each.

Organize teams around decisions and contracts; vendors are implementation choices.

01 HARDWARE	**Machines · network fabric · storage** NICo where authorized; provider and physical tooling	Qualified hosts and services Repair • isolation • sanitization
02 CAPACITY	**Atlas · Hydra · Kubernetes · Run:ai** GPU Operator/runtime • CNI • CSI • platform state	Exclusive host assignment Placement • drain • recovery
03 WORKLOAD	**Iris · Dynamo · vLLM · Croesus** Dedicated products • API/web/CLI • customer identity	Customer outcome and SLO Deployment • usage • ledger

Two hardware authority profiles

The same capacity interface must describe different powers and recovery guarantees.

FULL LIFECYCLE CONTROL

Lyceum controls provision, power, firmware, repair and sanitization.

NICo is a candidate below Atlas.
Hardware supplies a qualified base image.

HOST ACCESS ONLY

The provider controls physical lifecycle.
Lyceum configures only what access permits.

Atlas/Hydra can use the host; NICo cannot supply missing permissions.

Every host advertises its allowed operations

Identity + site + bootstrap privileges + maintenance terms + recovery + sanitization evidence.
Atlas/PDB protection cannot fence a provider’s independent power-off or withdrawal.

Dedicated inference and machine rental differ

All products keep their customer contract above the shared capacity interfaces.

SHARED INFERENCE

Iris API + admission
↓
Dynamo + vLLM
↓
Run:ai / Kubernetes

Model service shared by requests

DEDICATED INFERENCE

Iris reserved deployment
↓
Dynamo + vLLM
↓
Run:ai / Kubernetes

Dedicated replicas / declared isolation

MACHINE / VM RENTAL

Product API
↓
Hydra + Atlas
↓
Authorized physical provider

Exclusive host; outside shared pool

API / web / CLI → domain services **Usage facts → Croesus**

Croesus owns prices and the ledger. Run:ai reports resources; it does not invoice customers.

Remove duplicate authority, retain useful signals

The rule is one writer for each decision—not one program for every layer.

DECISION	OVERLAPPING SOFTWARE	TARGET OWNER / RULE
Deployment lifecycle	Legacy gateway ↔ Iris controller	Iris writes; gateway forwards
Replica count	Iris / KEDA / Planner / Knative	Fixed Iris/DGD count for MVP
GPU allocation	Run:ai ↔ extra scheduler / Execlet	One scheduler per disjoint pool
Host teardown	Hydra / Atlas / provider / NICo	One drain grant; one executor
Driver + runtime	Base image / Ansible / GPU Operator	One Capacity-owned installer
Commercial ledger	Runtime meters / Atlas / Run:ai	Croesus; deduplicated usage facts

Keep layered health: Atlas access → Kubernetes node → Dynamo graph → Iris endpoint.

Make the handoffs the team contracts

Each team operates its layer and publishes evidence the next layer can trust.

HARDWARE
Qualified host / storage / fabric

→

CAPACITY
Eligible cluster / resource class

→

WORKLOAD
Ready product / usage events

HostReady	Asset + authority profile + site/rack + image/trust + hardware health
CapacityClass	GPU/CPU topology + isolation + network/storage + limits + failure model
Maintenance	Workload SLO check → Capacity drain/verify → authorized physical action
UsageEvent	Durable source/event ID + quantity + provenance → Croesus pricing/ledger

Existing interfaces, versioned and tested. No new middleware service is implied.

One fleet across N main sites + P small sites

N ≥ 3 main sites. Each small site has tens of machines; distribution is still to be defined.

1,250

GPU hosts · 10,000 GPUs at 8 per host

<10,000

CPU hosts · includes management

1,000

Storage hosts · capacity / IOPS TBD

N MAIN SITES · N ≥ 3

Host mix and size can differ by datacenter.
Cluster count follows the local inventory.

P SMALL SITES · TENS OF HOSTS EACH

Use a separately sized operating profile.
Avoid copying the large-site footprint.

These are fleet totals across all N + P sites—not per-site allocations.

A site is not a cluster: define its GPU / CPU / storage mix before fixing cluster counts.
Both full lifecycle control and host-access-only sites fit this topology.

Budget control compute; measure the rest

Preliminary load-test allocations. VM resources below are not physical-server counts.

PER LARGE COMPUTE CELL

3 control VMs × 16 vCPU / 64 GiB
3 system VMs × 32 vCPU / 128 GiB

Total: 6 VMs · 144 vCPU · 576 GiB

EXAMPLE: 15 LARGE COMPUTE CELLS

90 VMs · 2,160 vCPU · 8,640 GiB

Illustrative subtotal, not the fleet total.
Actual count depends on site allocation.

Additional budgets and inputs—not hidden in the subtotal

N main sites + P separately sized small sites.
Management, DB HA, telemetry and APIs.
Reserve physical hosts inside the CPU cap.

Storage: usable bytes, IOPS, retention, rebuild.
Network: model-load bursts, RDMA needs.
Host-only sites: provider recovery contract.

Split ownership now; qualify scale in stages

Three accountable teams, with workload sub-owners as the organization grows.

HARDWARE

Own: machine/fabric/storage or provider contract.

Prove: host readiness, isolation, repair and return evidence.

CAPACITY

Own: Atlas/Hydra, Kubernetes, Run:ai and safe drains.

Prove: allocation, recovery and control SLO.

WORKLOAD

Own: Inference; Compute products; API/Commercial.

Prove: endpoint SLO, tenant isolation and correct billing.

MVP → site qualification → fleet qualification

AWS: 1 → 2 GPUs Site: 1 → 2 model hosts GPU cell: 25 → 100 → 250 hosts
Gate each step on failure, upgrade, restore and billing tests; size storage before buying it.